

Mayank Joshi

📞 8894267398 ✉ joshimayank08012007@gmail.com 🌐 github.com/lothnic 📁 mayankjoshi0801

Education

Indian Institute Of Information Technology, Una

Bachelor of Technology in Electronic And Communication Engineering (CGPA: 8.14)

Expected May 2028

Una, HP

Experience

Indian Institute Of Technology, Ropar | *Research Intern*

2026

- Researching LLM alignment using Direct Preference Optimization (DPO) and LoRA-based continual pre-training to mitigate catastrophic forgetting.
- Developing LLM-based low-resource Neural Machine Translation (NMT) systems for underrepresented languages.

Indian Institute Of Information Technology, Una | *Machine Learning Developer*

2025

- Built a genetic algorithm-based course scheduling system in Python, optimizing timetables across 50+ constraints to minimize scheduling conflicts.
- Integrated the optimization engine with the college website backend, enabling automated timetable generation.

Projects

Callisto – Open Source Agentic Browser | [GitHub](#) | Python, TypeScript, Playwright, CDP, LLMs, Browser Automation

- Built an open-source agentic browser enabling LLMs to autonomously browse the web, interact with dynamic pages, and execute multi-step workflows through a disciplined two-tool agent loop.
- Engineered a lean CDP wrapper in TypeScript (~15 methods, ~2K tokens) with a custom accessibility tree for token-efficient perception, plus a Python agent harness featuring browser state management, thrash/stuck recovery, budget tracking, and keychain-backed autofill.

Conflux: Civic-Tech AI for Urban Intelligence | [GitHub](#) | Python, FastAPI, Next.js, sentence-transformers, HDBSCAN, Leaflet

- Built an end-to-end civic AI platform ingesting multilingual complaints from Reddit and news RSS; applied LLM-based geocoding and geo-semantic HDBSCAN clustering over multilingual embeddings to surface infrastructure hotspots.
- Developed an agentic research pipeline using LLMs to generate structured proposals with INR budgets, agency assignments, and communication plans; streamed deep-dive analysis via SSE with policy RAG and Overpass POI context for downloadable planning reports.

Open-Source Contributions

- OpenHands (83k+ GitHub stars):** Recognized in the OpenHands Champions directory; shipped a Docker endpoint preflight check that surfaces UID and bind-mount permission failures at startup with actionable remediation, replacing a silent, partially initialized stack.
- Creator, VectorAlign:** Developed and published **VectorAlign**, a training-free bilingual word-alignment Python package using multilingual embeddings, supporting LaBSE/mBERT, bidirectional alignment, **batched GPU inference**, and efficient CUDA memory management.

Leadership & Technical Accomplishments

- ML Lead, IIITU IEEE Student Chapter:** Led technical workshops, seminars, and paper-reading sessions on Machine Learning and AI, fostering hands-on learning and research discussions among student members.
- Building vLLMini (Technical Article, July 2026):** Authored a technical deep-dive on **vllmini**, a minimalist LLM inference engine implementing PagedAttention and KV-cache management with custom CUDA kernels, exploring GPU memory optimization and efficient attention.
- Competitive Programming:** Solved 300+ DSA/CP problems across **LeetCode**, **Codeforces**, **HackerRank**, and **GeeksforGeeks** and achieved a **2-Star rating on CodeChef**, demonstrating strong algorithmic problem-solving skills.

Skills

Languages: Python, C/C++, Javascript, SQL, Bash

Backend & Distributed Systems: FastAPI, Rest APIs, PostgreSQL, Kafka Linux, AWS

Devops & Infrastrucure: Docker, Docker Compose, CI/CD Pipelines, Git, vLLM, HuggingFace

Deep Learning & CV: PyTorch, TensorFlow, OpenCV, Computer Vision, W&B, MLflow, NumPy, SciPy, Pandas

Tools & Frameworks: Cursor and Claude Code